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Evidence context

Distinguish association from causation and group research from individual diagnosis. Pair with the neurocircuit limitations review.

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Neuroimaging findings in obsessive-compulsive disorder: A narrative review to elucidate neurobiological underpinnings

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Abstract

Obsessive compulsive disorder (OCD) is a common psychiatric illness and significant research has been ongoing to understand its neurobiological basis. Neuroimaging studies right from the 1980s have revealed significant differences between OCD patients and healthy controls. Initial imaging findings showing hyperactivity in the prefrontal cortex (mainly orbitofrontal cortex), anterior cingulate cortex and caudate nucleus led to the postulation of the cortico-striato-thalamo-cortical (CSTC) model for the neurobiology of OCD. However, in the last two decades emerging evidence suggests the involvement of widespread associative networks, including regions of the parietal cortex, limbic areas (including amygdala) and cerebellum. This narrative review discusses findings from structural [Magnetic Resonance Imaging (MRI), Diffusion Tensor Imaging(DTI)], functional [(functional MRI (fMRI), Single photon emission computed tomography (SPECT), Positron emission tomography (PET), functional near-infrared spectroscopy (fNIRS)], combined structural and functional imaging studies and meta-analyses. Subsequently, we collate these findings to describe the neurobiology of OCD including CSTC circuit, limbic system, parietal cortex, cerebellum, default mode network and salience network. In future, neuroimaging may emerge as a valuable tool for personalised medicine in OCD treatment.

Keywords: Neurobiology, neuroimaging, obsessive-compulsive disorder

INTRODUCTION

Obsessive-compulsive disorder (OCD) is a common psychiatric illness characterized by recurrent, intrusive thoughts, images, or impulses and repetitive acts or mental rituals. Besides schizophrenia, it is one of the main psychiatric disorders wherein significant differences between patients and healthy controls have been observed from neuroimaging studies dating back to the 1980s. Based on the hyperactivity in the prefrontal cortex (PFC) (mainly orbitofrontal cortex [OFC]), anterior cingulate cortex (ACC), and caudate nucleus demonstrated in the initial studies, the cortico-striato-thalamo-cortical (CSTC) model of OCD neurobiology was postulated.[1] Most of the early studies focused on the CSTC loop; however, in the previous two decades, emerging evidence suggests

the involvement of widespread associative networks including regions of the parietal cortex, limbic areas, and cerebellum. In this narrative review, we discuss findings from structural (magnetic resonance imaging [MRI], diffusion tensor imaging [DTI]), functional (functional MRI [fMRI], single photon emission computed tomography [SPECT], positron emission tomography [PET], functional near-infrared spectroscopy [fNIRS]), and combined structural and functional imaging studies and subsequently collate these to describe the neurobiology of OCD. The study designs include (a) cross-sectional studies, (b) baseline and posttreatment imaging studies, and (c) symptom provocation paradigms comparing OCD patients and healthy controls. We have included data from only adult OCD patients for ease of understanding because child and adolescent imaging findings differ in a larger meta-analysis, suggesting neurodevelopment-related changes across age in OCD.[2]

STRUCTURAL IMAGING STUDIES (MAGNETIC RESONANCE IMAGING AND DIFFUSION TENSOR IMAGING)

MRI studies in OCD were initially region of interest based (investigate areas previously implicated in OCD), and later, voxel-based morphometry (VBM) was used (whole-brain analysis approach) for studying volumetric differences. Recent studies have used surface-based analysis to investigate cortical surface area and thickness. White matter (WM) changes have been studied using DTI. DTI is based on quantification of the diffusion of water molecules in the brain tissue and provides information about WM integrity and microstructure.[3] Fractional anisotropy (FA) is the main measure from DTI, and it depends on water molecule diffusivity which in turn depends on fiber density, axonal diameter, myelin sheath thickness, and fiber directionality. Decreased FA is suggestive of change in WM integrity.[3] A summary of MRI studies is shown in Table 1.

Table 1.

Magnetic resonance imaging studies in obsessive-compulsive disorder

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The largest mega-analytical study Enhancing NeuroImaging and Genetics by Meta-Analysis (ENIGMA) by pooling data of cortical thickness and surface area in 1905 patients and 1760 healthy controls showed lower surface area in the temporal cortex and thinner inferior parietal cortex, implicating areas other than those prevalent in the CSTC model.[2] Another mega-analysis comprising 412 OCD patients and 368 healthy controls showed similar findings of decreased parietal and temporal cortical thicknesses and cortical thinning in the right dorsolateral PFC (DLPFC), left posterior cingulate cortex (PCC), and bilateral hippocampi.[7] Other meta-analysis of VBM studies showed decreased gray matter (GM) in bilateral OFCs[4,5,6] and ACC[4,5,6] and increased GM in the basal ganglia (caudate,[4] putamen,[5] and pallidum[8]). Findings with regard to the parietal cortex were inconsistent with one meta-analysis showing increased volume[4] and the other showing decreased volume.[5] Increase in cerebellar GM was found in one meta-analysis.[6] In addition, investigation of age-related changes suggested relative preservation of putamen and insular GM and decrease in medial and lateral temporal cortical volume in OCD as compared to healthy controls with increasing age.[6] Differences with respect to symptom dimensions were reported in two mega-analyses.[6,7] Cortical thickness was increased in the left OFC[7] in contamination/cleaning, right OFC, left cingulate cortex, right parietal cortex,[7] and middle temporal cortex[6] in sexual/religious; right occipital[7] and lingual gyrus[6] in aggression/checking; and left insula, lingual, precentral, and postcentral gyrus[7] with decrease in fusiform GM[6] in symmetry/order symptom dimension. Most studies had recruited patients across heterogeneous symptom dimensions, contributory to the variability in imaging findings making comparisons difficult. In addition, other inclusion/exclusion criteria such as presence or absence of

comorbidity, medication status (drug-naive, drug-free >1 month), and varied ages of onset of symptoms are additional confounding factors. Association of symptom severity with imaging has also been investigated; some studies have shown no relation[6,15] while others have shown that Y-BOCS (Yale Brown Obsessive Compulsive Scale) severity scores have a positive correlation with increase in left dorsal ACC thickness,[11] decrease in cortical thickness in bilateral occipital gyri,[7] and negative correlation with GM volume of left ventral striatum,[14] left frontal pole volume, right anterior cingulate gyrus volume and surface area, and right OFC gyrification.[9]

Findings from DTI studies in OCD are shown in Table 2.

Table 2.

Diffusion tensor imaging studies and combined magnetic resonance imaging+diffusion tensor imaging studies in obsessive-compulsive disorder

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A meta-analysis of DTI studies in OCD revealed WM abnormalities in ACC and OFC,[33] similar to MRI studies. In DTI studies, positive correlations have been found between symptom severity, decreased FA in the right cingulum bundle[16] and forceps minor,[31] decreased axial diffusivity in the uncinate fasciculus, and increased FA in the left striatum[21] and left inferior parietal lobule;[25] whereas negative correlations with apparent diffusion coefficient of left insula, left lentiform nucleus, and left cingulate cortex.[25]

Among DTI studies, symptom dimension and symptom severity have been analyzed by Yagi et al., 2017;[28] checking severity showed negative correlation with left inferior frontal gyrus WM and middle temporal gyrus FA, and ordering severity showed positive correlation with the right precuneus FA and negative correlation with right precuneus axial diffusivity. Overall, obsessional severity showed negative correlation with axial diffusivity and mean diffusivity in the right supramarginal gyrus.[28] Kim et al., 2015[24] using a combined multimodal fusion analysis of structural MRI and DTI found that WM networks around the basal ganglia were associated with the symptom dimension of checking.

MAGNETIC RESONANCE SPECTROSCOPY

Proton magnetic resonance spectroscopy (1H-MRS) allows in vivo quantification of specific neurochemicals in different brain regions. Using magnetic field and a brief, tuned radiofrequency pulse, MRS generates resonance signals from hydrogen nuclei (protons) in neurochemical molecules, yielding a magnetic resonance spectrum; each molecule has a unique peak and the strength of each resonance reflects the molecule's concentration. N-acetyl aspartate (NAA), a marker of neuronal integrity, is the most widely reported neurochemical; others include creatine + phosphocreatine (total Cr), choline-containing compounds, myo-inositol, glutathione, lactate, and the amino acids - glutamate, glutamine, and γ-aminobutyric acid.[34]

MRS studies in OCD have targeted different parts of the CSTC circuit. Details of MRS studies in OCD are mentioned in Table 3.

Table 3.

Magnetic resonance spectroscopy and combined magnetic resonance spectroscopy+functional magnetic resonance imaging studies

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Studies targeting cortical regions have found increased NAA/Cr[38] and decreased glutamate[44] in the medial PFC, decreased glutamate and increased myoinositol in the DLPFC,[48] and decreased NAA in the ACC,[40,41] which increased after medication.[40] An older study had slightly different findings, with lower NAA concentration being found in the ACC in responders to selective serotonin reuptake inhibitors (SSRIs) with antipsychotic augmentation as compared to healthy controls, while no such differences were seen between SSRI responders and nonresponders.[36]

Studies targeting the striatum found decreased NAA levels.[35,41] A pre- and post-medication study found decreased NAA/Cr levels in the ACC and caudate in OCD patients at baseline, which normalized after 12 weeks of medication.[42] Similar results were seen in the caudate nucleus after 12 weeks cognitive behaviour therapy.[39] However, negative results with no difference in striatum between OCD patients and healthy controls were found in another study.[45]

A few studies targeting the thalamus[44,47] found decreased glutamate levels in the right thalamus[44] and increased choline levels in the midline in bilateral thalamic regions.[47] Both studies showed a positive association of glutamate with Y-BOCS symptom severity.[44,47] Among the other regions studied, Atmaca et al., 2009 found decreased NAA/Cr in the hippocampus.[37]

Brennan et al., 2015 reported increased activation in the rostral ACC on fMRI during emotional counting Stroop paradigm; however, no glutamate/metabolite abnormalities were noted in the ACC on MRS.[46] A meta-analysis of 17 MRS studies in OCD found decreased NAA levels in the frontal cortex, but no significant change in the basal ganglia. Meta-regression revealed that NAA reduction in the medial PFC positively correlated with Y-BOCS symptom severity.[49]

Two studies that combined DTI and MRS targeting the dorsal ACC and anterior thalamic radiation (ATR) found positive correlation between choline levels and FA in dorsal ACC and negative correlation between choline levels and fiber length in the right ATR[50,51] as shown in Table 4. In addition, FA in the ACC and right ATR correlated with Y-BOCS symptom severity.[50,51] The summary of combined structural and functional imaging studies (DTI + MRS) is shown in Table 4.

Table 4.

Combined structural and functional studies

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FUNCTIONAL IMAGING

Functional imaging studies measure neural activity at rest (PET, SPECT, and fMRI) and during cognitive tasks (fMRI and fNIRS).

Positron emission tomography and single photon emission computed tomography

PET studies measure regional glucose metabolic rates which correlate with brain activity. Since the early 1980s, studies using PET have shown areas of hypermetabolism in OCD in different regions of the PFC.

Increase in metabolism has been reported in the left OFC,[56,57,58] bilateral ACC,[57,59] bilateral caudate,[56,60] left premotor cortex,[61] right caudate,[62] putamen,[59,60] and thalamus[59] and decreased metabolism in the DLPFC.[58] Symptom severity associations have been reported with hypermetabolism in the left

OFC, bilateral PFC, and ACC.[57]

Posttreatment reduction in metabolism has been reported in the right OFC,[63] cingulate cortex,[59] and right caudate.[60,63,64] However, another study has reported increase in caudate metabolism following successful treatment with paroxetine or CBT.[65] This difference in the finding could be due to improvement in depressive symptoms as assessed on Beck's Depression Inventory in the OCD patients rather than an isolated reflection of change in OC symptoms.[65]

SPECT studies focus on blood flow, receptor availability, and drug uptake. SPECT studies in OCD have found reduced uptake in the OFC, ACC,[66] PCC[67] and temporal, parietal, and occipital cortices[67] and increased uptake in the cerebellum.[66]

Receptor availability studies have shown decreased striatal dopamine transporter[68] and decreased thalamic, hypothalamic,[68,69] and midbrain serotonin transporter.[68,70] However, another study showed increased midbrain serotonin transporter which was more pronounced in early-onset OCD.[71]

With respect to Y-BOCS symptom severity, positive correlations have been found with OFC,[66,72] DLPFC, lateral and medial temporal cortex, and inferior parietal lobule uptakes[72] and negative correlations with posterior cingulate uptake[66] and serotonin transporter availability in hypothalamic and thalamic areas.[68,69]

A recent meta-analysis of eight PET and six SPECT studies, which included 188 OCD patients in a pre- and post-treatment design, found that decrease in metabolism in the caudate, OFC, and thalamus on PET and decrease in blood flow in the caudate on SPECT were associated with improvement corresponding to normalization of the CSTC circuit overactivity.[73] A summary of PET and SPECT studies is provided in Table 5.

Table 5.

Positron emission tomography and single photon emission computed tomography studies

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Despite a large number of PET and SPECT studies available, they had limitations such as small sample sizes and differences in inclusion/exclusion criteria with respect to comorbidity, heterogeneous symptom dimensions, and medication status. Moreover, the reference region for measurement varied in the SPECT studies, with some using cerebellum[69,70] and others using occipital cortex.[71] Due to these limitations, it is difficult to generalize the findings. However, many studies have investigated pre- and post-treatment changes, rendering them a useful tool to understand the neurobiology of OCD including identifying regions for treatment modulation.

Functional magnetic resonance imaging

fMRI is based on blood oxygen level dependence response in different brain regions and assessment of changes in activation and connectivity. fMRI acquisition is commonly done in the resting state, during symptom provocation, and along with cognitive tasks. There are two types of resting-state fMRI studies in OCD: (a) hypothesis-driven, seed-based analyses, based on local abnormalities mainly focusing on the frontostriatal circuit, and (b) hypothesis-free, data-driven analyses, based on global abnormalities. A summary of fMRI studies is shown in Table 6.

Table 6.

Functional magnetic resonance imaging studies

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Resting-state fMRI studies have found increased functional connectivity between the caudate, putamen, and OFC, and ACC and parahippocampal areas.[88,90,92,93,100,103,104] In addition, the occipital cortex,[95,100] cerebellum,[95,100] and thalamic[90] connectivity with the striatum has been found to be increased. Decreased functional connectivity has been found within the OFC,[95] cerebellum,[94] and occipital areas.[94] Decreased functional connectivity has also been found between the dorsal striatum and lateral PFC,[88] ventral striatum and midbrain,[88] and posterior temporal region.[90,94] Symptom severity has been positively associated with connectivity in the OFC,[92,100] ACC,[104] putamen,[92,95] cerebellar region,[95] and lower dorsomedial PFC (DMPFC) intrasystem connectivity and the connectivity of DMPFC with striatal and cingulate areas.[96]

A symptom provocation fMRI study revealed greater activation in the ventromedial PFC (VMPFC) and caudate in washers; putamen, thalamus, and dorsal cortical areas in checkers; and left precentral gyrus and right OFC in hoarders.[85] Symptom provocation tasks and emotional processing tasks have found amygdala activation in OCD[91,97,98] with aggression/checking and sexual/religious symptom dimensions showing maximum correlation.[97]

Studies comparing activation on cognitive tasks in OCD patients and healthy controls have revealed differences in CSTC circuits as well as cerebellum and parietal areas. On the Stroop task, OCD patients showed decreased activation in the right ACC,[86,110] right caudate,[86] and right cerebellum.[110] On the n-back working memory task, increased activation in the right DLPFC, left superior temporal gyrus, left insula, and cuneus[87] and reduced activation in the right inferior parietal lobule and left supplementary motor area (SMA)[102] have been observed. On the Tower of London task, decreased activation of the right DLPFC and decreased connectivity between DLPFC and putamen were seen in OCD compared to healthy controls.[101] Symptom severity was found to be associated with right OFC activity[87] and SMA modulation.[102]

Newer fMRI studies have measured the amplitude of low-frequency brain fluctuations (aLFF) in various brain regions in OCD[30,106,107] and found the aLFF to be decreased in the OFC[107,111] and occipital and parietal regions[111] and increased in DMPFC[107] and temporal regions.[111]

A recent meta-analysis of 18 whole-brain resting-state fMRI studies with 541 patients and 572 healthy controls compared functional connectivity. In OCD, decreased connectivity within the frontoparietal and salience networks; between the salience, frontoparietal, and default-mode networks; and general dysconnectivity (no specific increase/decrease of connectivity) within the default-mode, frontoparietal, and salience networks were found.[112]

Pre- and post-medication-related fMRI studies have been summarized in Table 7.

Table 7.

Functional magnetic resonance imaging studies pre- and post-treatment

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These have shown postmedication increase in frontoparietal connectivity;[114] decrease in connectivity of the ventral striatum;[92] and decrease in activation of the ACC, SMA, DLPFC, and parietal cortex on working memory task.[113] Predictors of improvement with medication included pretreatment activation of the left superior temporal cortex and right cerebellum on symptom provocation task.[89] Studies investigating fMRI changes post-CBT have

found improved connectivity between the cerebellum and widespread areas in the caudate, putamen, frontopolar cortex, DLPFC, and VLPFC[115] and normalization of increased left DLPFC and right OFC connectivity.[116] Predictors of improvement following CBT included baseline decreased basolateral amygdala (BLA)-VMPFC functional connectivity.[105]

Functional near infra-red spectroscopy

NIRS is a new optical imaging modality which is portable and easy to use. It is based on the principle of neurovascular coupling and measures oxygenated and deoxygenated hemoglobin levels in brain areas to determine brain activity.[117] In fNIRS, NIRS is combined with cognitive/behavioral tasks. Two fNIRS-based studies in 20 and 12 adult OCD patients have been published.[118,119] These found decreased activation in the DLPFC during the verbal fluency task[118] and in the left lateral PFC on Stroop task[119] in OCD patients compared to healthy controls. A single case report has shown increased activation in the frontal and temporal areas on the verbal fluency task following improvement with escitalopram and CBT.[120] These studies are in line with studies in other modalities of imaging, implicating PFC areas in the neurobiology of OCD and their modulation, leading to improvement.

TOWARD UNDERSTANDING NEUROBIOLOGY FROM NEUROIMAGING

Neuroimaging findings dating back to the 1980s led to the initial understanding of the neurobiological basis of OCD. The CSTC model was proposed by Saxena et al., 1998.[1] It consists of an orbitofrontal loop, with projections from the OFC to the head of caudate nucleus and ventral striatum, connecting via the globus pallidus interna to the mediodorsal thalamus, and back from the thalamus to the OFC completing the loop. In addition, Saxena et al. proposed that OCD is mediated by an imbalance between the direct (excitatory, OFC-striatum-globus pallidus-thalamus-cortical) and indirect (inhibitory, DLPFC-striatum-globus pallidus-subthalamic nucleus-cortical) pathways within this circuit, which leads to hyperactivation across the OFC and thalamus.[1] Structural imaging studies supported the CSTC model with meta-analyses showing GM increase in the caudate and putamen[4,5,16] and decreased GM in the OFC.[4,5,16] Abnormalities in fractional anisotropy (reflecting white matter integrity) in the OFC,[25] ACC,[20,29] PFC,[22,24,25] and striatum[21] have been found in individual studies and meta-analyses.[33] In addition, functional studies have supported the CSTC model with a recent meta-analysis of PET and SPECT studies showing a decrease in metabolism in the caudate, OFC, and thalamus on PET and decrease in blood flow in the caudate on SPECT to be associated with improvement corresponding to normalization of the CSTC circuit overactivity.[73]

However, imaging findings have also implicated regions outside the CSTC loop, such as the parietal cortex, occipital cortex, and cerebellum in both structural[2,4,5,6,7,16] and functional studies; modified models of this circuit have been suggested postulated to include limbic regions (representative of the affective domain) connected to the OFC: hippocampus, anterior cingulate, and BLA.[58,66,88,91] Posterior brain regions, namely parietal, occipital, and cerebellar cortices (representative of the visuospatial domain) which are part of the posterior attention system involved in disengaging spatial attention, have also been implicated in OCD.[24,53]

The role of limbic system in OCD particularly the amygdala has been seen in neuroimaging studies.[46,52,53] Amygdala role in anxiety and negative affect has been well established and its association with compulsivity through connections with the corticostriatal system.[121] Recent studies show increased amygdala reactivity to emotional face stimuli and additionally increased activity in frontoparietal, OFCs, thalamus, and insula.[91,97,98] Findings from these studies suggest the limbic system activity is not altered during rest, but emotional stimuli lead

to activation during exposure to OCD triggers.[122]

The other areas of limbic system overlap with default mode network (DMN) (comprising the posterior cingulate cortex, medial PFC, and lateral parietal lobule) and the salience network (SN) (comprising the dorsal ACC and anterior insular cortices). DMN and SN have been found to have reduced interconnectivity in OCD. These two networks have neural activity that is inversely correlated, and attentional processes can normally shift from internally focused introspective states mediated by the DMN to externally focused activity mediated by the SN.[123] In OCD, due to reduced inverse connectivity between the DMN and SN, there could be failure to de-active the DMN; subsequently, the introspective functioning may intrude upon the focused functioning required of the SN.[109] In addition, within the DMN, imaging studies have found increased PCC-VMPFC and decreased PCC-ACC and putamen functional connectivity in OCD.[96,124] PCC activity increases during attention to targets that are of high motivational value.[125] OCD patients may have increased responsiveness to symptom relevant stimuli due to a combination of increased activity in the direct excitatory CSTC pathway and decreased PCC activity in response to other rewarding stimuli, leading to increased activity with the DMN itself and subsequently greater focus on internal states and inability to suppress intrusive thoughts.[108] Dysfunction of posterior brain regions, especially the cerebellum, contributes to the pathogenesis of OCD through impaired inhibitory control; normalization of cerebellar function can occur with an improvement in OCD symptoms.

Our understanding of the neurobiology of OCD and its evolution over a neurodevelopmental perspective is expanding with ENIGMA consortium working group collecting data from 16 countries to generate neuroimaging meta-analyses of nearly 2000 patients across all age groups.[2,8] Such large-scale multicentric studies help in overcoming limitations of existing neuroimaging studies including small sample sizes and differences in inclusion/exclusion criteria with respect to comorbidity, heterogeneous symptom dimensions, and medication status and enable comparison of data. There is potential for neuroimaging evolve as a potential marker of response as well as illness phenotype.

Newer approaches involving machine-learning algorithms in neuroimaging may provide better insights into the future and develop personalized medicine. A recent study by Reggente et al., 2018 leveraged machine learning to assess pretreatment functional connectivity patterns within the default mode network and visual network and found these significantly predicted posttreatment OCD severity.[126] Further studies using machine learning in imaging are ongoing, and greater predictive power and understanding of neurobiology are likely in the near future.

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Conflicts of interest

There are no conflicts of interest.

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